# Exercise 11

## Coyote time

At the end, a jump pressed a moment after running off an edge still counts.

## Before you start

- Exercise 10 done and pushed: a tap gives a short hop. In a new session, run `git pull` first

## 1 Run 6: coyote time

Players often press jump a moment after running off an edge, and a refused jump feels like the game's fault. Coyote time still allows the jump for 0.1 s after the walker leaves the ground, like a cartoon coyote that runs past a cliff's edge and hangs in the air for a moment before it falls.

Below `groundLayer`, add:

```
    [SerializeField] private float coyoteTime = 0.1f;
    private float coyoteTimer;
```

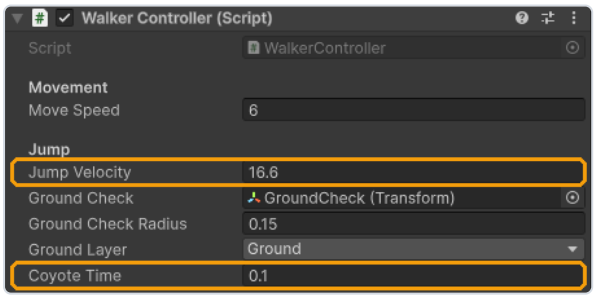
In `FixedUpdate`, after the `OverlapCircle` line, add the timer, full while grounded and counting down in the air:

```
        if (isGrounded) coyoteTimer = coyoteTime;
        else            coyoteTimer -= Time.fixedDeltaTime;
```

Make the jump ask the timer instead of `isGrounded`, and empty the timer when the jump happens, so the window can't give a second jump:

```
        if (jumpRequested && coyoteTimer > 0f)
        {
            rb.linearVelocity = new Vector2(rb.linearVelocity.x, jumpVelocity);
            coyoteTimer = 0f;   // a granted jump spends the whole window
        }
```

- **Play** and jump just past the ledge's edge with Coyote Time at `0`, `0.1` and `0.5`: late presses are refused, then accepted, then the walker jumps in mid-air. Values changed in Play mode go back to `0.1` when you stop.




The walker's settings at the end: Jump Velocity 16.6 and Coyote Time 0.1.

- `Ctrl` + `S`, then commit and push with the message `Add coyote time`.